

Enhancing PAN Aadhaar Linkage

Epic / Story	 MCA-820: [PAN Profile] [App] Enhance PAN Profile Implemenation  MCGBE-1235: [PAN Profile] Enhance PAN Profile Implementation  MCGFE-346 Can't find link
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Context

As part of our fraud check mechanisms, we currently use PAN Profile API from Hyperverge and Karza which helps us validate if the Aadhaar Number and PAN are linked to each other or not. Basis our current run rate as of Apr'23, we do around 2 lakh+ verification via external service providers.

We currently do this while moving the app to NACH for all customers, and we get a boolean response on

- whether the pan and aadhaar linked match the information we have and
- whether the date of birth on the pan matches the date of birth we have.

The API hasn't been working for the last 5+ days (30th May - Till Date) and it poses a significant risk from a fraud standpoint, therefore we are looking for alternates.

Older PRD for reference - [Implementing PAN Linked Aadhaar Matching through Karza API](#)

Objective

The objective is to scrape the open Aadhaar PAN linkage check URL and utilize the API directly to determine if the PAN is linked to the same aadhaar or not.

We need to do this for all types of KYCs.

This is also a cost-saving initiative as we can eliminate the usage of PAN Profile from third-party vendors (Karza/HV) altogether.

 Full Aadhaar Number should never be stored in any of our databases.

🤔 Assumptions/Considerations

For the sake of this PRD, we will call the new service as ITD Service.

⚠️ Out of Scope

Users on the older versions of the app will be out of scope.

Partners

ITD EndPoint for PAN Aadhaar Check

Aadhaar PAN Linkage Check → [e-Filing Home Page, Income Tax Department, Government of India](#)

Sample Responses -

Scenario	Response	Interpretation
Invalid	PAN does not exist.	pan_valid = false
Invalid	Please enter a valid Aadhaar Number.	aadhaar_valid = false
Linked to Other Combination	Your PAN AIXXXXXXX9G is linked to some other Aadhaar 90XXXXXXXXX45. To delink kindly submit request to your Jurisdictional Assessing Officer.	pan_valid = true aadhaar_valid = true pan = AIXXXXXXX9G aadhaar = 90XXXXXXXXX45 link = true match = false
Linked to Other Combination	Your Aadhaar Number 58XXXXXXXXX74 is linked to some other PAN BXXXXXXXX3M.	pan_valid = true aadhaar_valid = true pan = BXXXXXXXX3M aadhaar = 58XXXXXXXXX74 link = true match = false
Linked to Correct Combination	Your PAN ANXXXXXX4D is already linked to given Aadhaar 51XXXXXXXXX10	pan_valid = true aadhaar_valid = true pan = ANXXXXXX4D aadhaar = 51XXXXXXXXX10

		link = true match = true
Not Linked	PAN not linked with Aadhaar. Please click on Link Aadhaar link to link your Aadhaar with PAN.	pan_valid = true aadhaar_valid = true link = false match = null

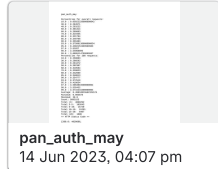
We should save the response of this API for future purposes. The exact string is to be saved as a remark for debug purposes.

These strings and error codes should be implemented based on an intelligent NLP-based algorithm to accommodate for any changes in the remarks. Keep any relevant changes required in the property file.

Karza PAN Authentication

API - <https://testapi.karza.in/v2/pan-authentication>

API Performance Data from Karza -



Sample Request -

```

1 {
2   "pan": "BXXXXXXXXR",
3   "name": "Omkar Milind Shirhatti",
4   "dob": "17/08/1987",
5   "consent": "Y"
6 }
```

Sample Response -

```

1 {
2   "result": {
3     "status": "Active",
4     "duplicate": null,
5     "nameMatch": true,
6     "dobMatch": true
7   },
8   "request_id": "deff5ed8-0460-11e9-a082-4742912ca12a",
9   "status-code": "101"
```

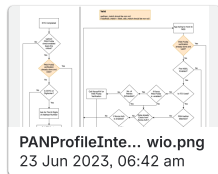
Design

App:  MVL 1.5V2 UI Refresh May

Web:  Web

Requirements

Flow



Whenever a user successfully completes KYC, we should

- take the aadhaar number from OKYC (or ask for aadhaar in case of CKYC or Digilocker)
- and PAN number on the loan app
- and use the ITD Service to validate if the two are linked to each other or not.

This should be done by the client or front end and the response should be posted to the back end to interpret and process it.

When moving the app to NACH, we should check if ITD Service was used. If yes, we should use it in further flow. Otherwise, we should fall back to the existing setup which uses Karza / Hyperverge.

- If linkage information exists from the ITD service.
 - If the information exists and Aadhaar PAN linked is false, we should use Karza API to fetch the DOB match information and persist accordingly.
 - If the information exists and Aadhaar PAN linked is true, we can directly rely on the match information, and no DOB call is needed.
- If linkage information **doesn't** exist from the ITD service, we should fall back to the existing setup which uses Karza / Hyperverge.

Overall, ITD service along with Karza PAN DOB check can be looked at as an alternative to the existing Karza and HV PAN Profile Service.

Requirements

Client should save the aadhaar number in the cache for the particular attempt only so that it can be used at the end of KYC completion. If the KYC fails, and user starts a new attempt the previously aadhaar number should be forgotten.

OKYC - Once the user completes KYC, client on indication from the back end should call the ITD service to validate the information and post the response to back end.

Digilocker - Once the user completes Digilocker, client on indication from the back end should ask for aadhaar number and call the ITD service to validate the information and post the response to back end.

CKYC - If the user is tagged to CKYC, client on indication from the back end should ask for aadhaar number and call the ITD service to validate the information and post the response to back end.

Back end should parse the response basis the configuration and validate that the PAN and Aadhaar Number are same on the loan app. Post validation, response can be saved in NSDL_PAN_Detail table.

At FR → NACH (Non-Seamless) / Submit → NACH (Seamless)

We should first check if the PAN Aadhar Verification is completed successfully using the internal service. If not we should hit the external API to determine if the PAN and Aadhaar are linked to each other. If yes, the PAN Profile API call should be skipped for such cases.

 We should make sure that Full Aadhaar Number is never stored in any of our databases.

KYC Resubmit

In the case of KYC Resubmit we should check if the PAN Aadhaar verification has been done already or not. If yes, we should not do it, otherwise, we should.

Also, for Resubmit, post KYC completion in the Resubmit flow, **we should check if both KYCs were completed on the same aadhaar number** by matching the last 4 digits, failing which the app will be rejected.

Reuse Logic

- Aadhar_match is NON NULL
Resue within 12 months if aadhar_match is true
Resue within 3 months if aadhaar_match is false and dob_match is NON NULL
- Aadhaar_match is NULL
Reuse within a month if dob_match is NON NULL

Topup and Repeat Cases

For Topup and Repeat apps, we have a reuse logic built in place. Same reuse logic would apply here.

- If the top-up or repeat app is within the defined reuse timeframe, the same service response should be used whichever was used for the original app.
- If the top-up or repeat app is outside the defined reuse timeframe, then the flow should be similar to a new app as defined above.
 - **Topup apps who need KYC but don't need PAN** - In this case we would have the PAN Profile data and should not ask for an Aadhar number.

Failures and Retries

- If the ITD Service fails with 4xx and 5xx errors, we should try immediately x times, if still not successful, we should let the customer move ahead.
- If the response from ITD Service contains an Error code that we do not have a mapping for, we should try immediately x times, if still not successful, we should let the customer move ahead.

 Even the failures like 4xx, 5xx should be posted to back end for us to monitor on the overall performance of the service.

Alerts

- **From Code** - Alerts should be sent on email and slack channel when the ITD service response parsing fails or the error code received is the response is not found mapped properly.
- **Via Metabase** - Proper alerting and monitoring mechanism (via Dashboard) to be created for all APIs so that we are aware of what's happening in the system. @Vineeta Verma

Identification

There should be a tag or a mechanism to identify the apps which went through the ITD Service flow.

Atlas

We need to identify whenever the KYC is being done via Atlas, and in such cases we should not give the ITD Service option. @Sahil Ahuja I hope this is possible.

Config

New Configs -

- **ITD Service Rollout %** - Meaning for what % is the ITD Service enabled. This also means collecting Aadhaar Number on Front End for CKYC and Digilocker apps in this case. This should be set on a percentage basis.
 - **Enabled Channel** - There should be a config through which we can disable this for KYCs done on Atlas. @Sahil Ahuja
- **No. of Attempts on Failure** - This defines how many times should the client try calling ITD Service in case of failure.
- **API Specific Flags**

Existing Configs -

- **Split between Karza and HV** - In case the check has not been done using ITD Service, existing config to determine the traffic split between Karza and HV should be used.
- `fire.nsd1.pan.detail.active` - true value of this prop would mean that we should be able to skip the overall NSDL check (including the ITD Service) while moving the app to NACH.

Phase 2 Thoughts

- We can reject the app right away at KYC once we know that the Aadhar PAN match is false.

- We need to build a flow to redirect the customer to the URL rather than our own page

Metrics

Success Metric

No Funnel Impact with Better Fraud Coverage -

- No Drop in KYC Done → Submits
- Same % coverage for PAN Profile as before

Other Metrics

Track the distribution of ITD Responses -

1. No. of “Invalid PAN” cases
2. No. of “Invalid Aadhaar” cases

Since this is also a cost-saving measure, we would measure the absolute number and the percentage of external requests saved with Karza and Hyperverge which will help in determining the exact cost savings.

Check Metrics

We will basically have to make sure that there are no issues in the flow with this new implementation and no additional users are getting REJECTED or are stuck in the funnel. The check metrics would include -

1. Funnel Conversions from KYC till NACH
2. Percentage and Absolute number of rejections post Submit due to PAN Aadhaar Not Linked

Monitoring Metrics via Dashboard

A dashboard to track the response split and success rate for the ITD API on a daily and hourly basis is to be created.

Alerts on API downtime and any significant change in the response reasons should also be set up with the threshold so that we are always informed and take action accordingly if needed.

Release

This will be a one-time release for all customers. Versioning or A-B functionality is not required.

Open Questions

Reporter	Question	Answer	Date Answered
Ishan	What happens in case of topups and repeat? Anything needs to be changed here?	Repeat and Topups logic and flow is added in the PRD	17/4/2023
Ishan	Consent required?	No	